

Ecological Methods

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Ecological studies involve setting objectives and formulating a sampling programme; sampling in the field and sorting, identifying and measuring the organisms retrieved; and data analysis. The results can be used to derive absolute or relative population measures, construct life tables, probe population dynamics and estimate biodiversity.

Introduction

The study of plant and animal populations has required the development of a large body of methods that use techniques derived from mathematics, physics, chemistry, biochemistry and systematics, combined with the ancient skills of the hunter and fisherman. The particular methods required will be determined by the objectives of the study and these must be clearly defined at the outset. Although it is common to divide the ecological methods used by botanists from those of zoologists, they share many common features and both will be considered here; the techniques used to study sedentary animals such as barnacles will be closer to those of the botanist than of the ornithologist. Studies can be broadly classified as intensive or extensive. Extensive studies are carried out over larger areas than intensive studies and seek to obtain information on the distribution and abundance of species for conservation and management purposes. Intensive studies involve the repeated observation of a population to gain insight into demographic processes.

All ecological studies follow a sequence of steps. First the objectives are defined and a sampling programme is formulated. Second, the field sampling is undertaken, the organisms are sorted and identified, and relevant measurements such as size or age are recorded. Finally, the data is tabulated and analysed, usually using a computer, and the results are reported. Care is required at all stages to ensure that the study is realistic given the manpower and other resources available.

Detailed accounts of the methods available to ecologists are given in Southwood and Henderson (2000) and the numerical methods are described in Krebs (1999) and Legendre and Legendre (1998).

Absolute and Relative Measures of Populations

Absolute estimates express population size in numbers per unit area or volume. Often it is more appropriate to count the number of animals per unit of habitat, e.g. per leaf or

per host. Such measures are termed population intensity and are more meaningful than absolute number when the objective is an assessment of crop or host damage. Relative methods produce some measure of abundance based on the number caught per unit of effort or trapping. Relative methods are frequently used when resources are limited, when it is desired to quickly obtain a general sample of the species present, or when the organism is difficult to sample quantitatively. Given constant sampling efficiency, relative methods can be used to follow population trends or compare abundance between habitats. Both absolute and relative methods will be discussed below.

The Sampling Programme

Without a well-designed sampling programme, an ecological study will likely fail. As it is not usually possible to count all the animals or plants in a population, samples must be taken and the population has to be estimated. The design of this sampling programme is a key ecological skill and will require decisions about the subdivision of the habitat, the size of the sampling unit, the number of samples, the spatial pattern of the samples and the time of sampling. All ecosystems are seasonal and it is important to identify the appropriate time of year when the life stages of interest are present. Knowledge about the pattern of distribution of the population is particularly relevant. While some organisms may be randomly distributed, highly aggregated patterns are far more common and this must be taken into account when deciding how to sample. A preliminary survey may be necessary before the final sampling design is formulated.

Absolute Population Estimates

The number of individuals in a fixed unit of habitat must be counted or the population studied by mark–recapture, removal sampling or distance sampling methods.

Secondary article

Article Contents

- Introduction
- Absolute and Relative Measures of Populations
- The Sampling Programme
- Absolute Population Estimates
- Relative Methods of Population Estimation
- Life Tables and Key Factor Analysis
- Species Richness and Diversity
- Recognizing Community Structure

Sampling the air

Suction traps are frequently used to sample small flying insects such as aphids from a known volume of air. Suction samplers are also used to sample pollen and airborne spores. These devices usually use an electric fan to pull or push air through a collecting filter. The amount of wind may greatly alter the efficiency of these sampling devices.

Sampling from herbage and trees

Powerful suction samplers that work on the same principle as vacuum cleaners are used to sample invertebrates, particularly spiders and insects, from low herbage such as grass. Arboreal insects are frequently sampled by chemical knock-down. The tree is sprayed with an insecticide such as pyrethrin and the killed insects are collected on sheets under the tree or branch. Knock-down techniques have been important for the estimation of tropical forest biodiversity.

One of the simplest and still widely used methods for short vegetation and open habitats such as sand and mud is quadrat sampling. A wire frame is positioned at random and the number of organisms within the area is counted. Botanists also use point sampling methods in which a fine point is lowered into the herbage and the leaves into which it comes into contact are recorded.

Sampling from aquatic habitats

Planktonic organisms are usually sampled with a plankton net that may be fitted with a flow meter to calculate the volume of water filtered. The size of mesh is determined by the organisms of interest, but should be as large as possible, since clogging is a frequent problem. Pump and box samplers are also used in habitats where nets are unsuitable. The fauna on the bed of a stream is usually sampled using a kick sampler, which is a modified net that catches the animals washed from the gravel following disturbance. The bottoms of lakes, deeper rivers and marine habitats are sampled by a variety of dredges, trawls, grabs and corers. Grab and core samplers are the only techniques able to give reliable absolute population estimates as they cut out a known area of seabed that can be lifted to the surface, where it is sieved and the organisms are extracted. Grabs are the method of choice for mud and sand substrates in marine environments and a large number of designs are commercially available. Corers are frequently used on intertidal habitats and the beds of lakes. These may be as simple as a piece of plastic pipe that is pushed into the mud.

Sampling from soil and litter

Samples are usually taken with corers. The main problem facing the ecologist in this habitat is the extraction of the

organisms from the matrix, and a wide variety of techniques can be used. It may be possible to sieve either wet or dry, but, for many invertebrates, dynamic methods such as the Berlese–Tullgren funnel are used. The sample is placed in the apparatus and the animals are driven out by drying or heat.

Mark–recapture

This technique is mainly used by animal ecologists, as it requires individuals released back into the population to quickly become randomly distributed. The basic idea is to capture a known number of individuals, mark them for future recognition and release them back into the population. A subsequent sample is taken and the proportion of this sample that is marked then allows the calculation of the total population size. A wide range of approaches have been developed that require a series of samples and allow migration, birth and deaths to be taken into account. A key requirement of these techniques is that the capture and marking does not harm the animal or alter the behaviour. Marks can take a variety of forms including dots of paint, fluorescent powder, tags and branding. The principal limitation of these techniques lies in the large proportion of the population that must be marked (often more than 20%) if a reliable population estimate is to be obtained. Other problems arise from the differential catchability of animals within populations.

Removal trapping

This approach can only be used on a population that is closed to migration. It has the advantage that it can use relative sampling methods such as trapping to estimate population size. The basic approach is to repeatedly sample or search the area and record the number found and removed on each occasion. Providing that the number found declines through time, the numbers caught can be used to estimate the original number present. This technique is commonly used to estimate fish densities by electric fishing. A reach of stream is netted off and fished three or more times. Removal trapping is essential in this case because electric fishing is of variable efficiency and never 100% efficient. Another possible application would be the estimation of ground beetle numbers by pitfall trapping. An area of ground is fenced off and pitfalls are used to gradually remove the beetles present.

Distance sampling methods

A wide variety of methods are included within this category. They are extensively used by botanists and ornithologists.

In closest-individual methods a point is selected at random and then the area around it is searched in

concentric rings; the closest individual is found and its distance from the point is measured. The process should be continued to find the 2nd, 3rd, ... i th closest individual. These measurements are then used to calculate an estimate of the density.

A related technique is nearest-neighbour estimation, in which an individual is selected at random and the search is continued until the nearest neighbour is encountered. The distance between neighbours is then used to calculate density. A wide variety of techniques based on this method are used, which may be more robust to nonrandom distributions. These include angle-order methods in which the distance to the nearest neighbour in each quadrat is found.

Point and line survey techniques are widely used for estimating the density of birds, butterflies and whales, and can be applied to large plants such as trees. To obtain estimates of population density, a function is calculated that describes the decline in the probability that an organism will be spotted with distance from the point or line. This function is then used to calculate the total population density. A commonly used function is based on the Fourier series.

Relative Methods of Population Estimation

An extremely wide range of techniques, from the most primitive trapping methods such as a pitfall to sophisticated electronic methods such as sonar or radar, will give relative population estimates. Traps are frequently used by animal ecologists and include light traps for night-flying insects and larval fish, pitfall traps for ground-living animals such as spiders and beetles, malaise traps for day-flying insects, traps and tangle nets for fish, and sticky traps. Traps, like most relative methods, depend on the activity of the animal and will give misleading information if some part of the population is inactive. For example, light traps usually have a variable and unknown level of efficiency. They may catch different numbers of moths depending on the weather and the state of the moon. Further, they may only catch one sex or a limited age range.

Radar is used to monitor bird migrations and even to follow the flight of large insects such as bees. Sonar survey methods are widely used to study fish distribution and the most sophisticated systems are able to estimate the number of fish in rivers or lakes. These systems work by recognizing the distinctive signal produced by the swimbladder of the fish.

Relative methods are widely used because they frequently give impressive species lists for modest effort. A full account of these methods is given in Southwood and Henderson (2000).

The use of signs, products and effects

The presence of some animals, particularly shy mammals and birds, is sometimes detected by surveying for footprints, faeces, hair, nests, burrows or cast skins. Signs, such as insect damage to plants, are also frequently used by economic entomologists.

Life Tables and Key Factor Analysis

In intensive studies, the construction of life tables is important for the understanding of species population dynamics. The life table shows the pattern of births and deaths through time. An age-specific life table is based on the fate of a real cohort, whereas a time-specific life table is based on the age structure of the population at some instant in time. The analysis of life tables by methods such as key factor analysis is used to identify particular stages in the life cycle that are particularly important in determining population size. Ecologists frequently seek to identify processes that result in density-dependent births and deaths. In a density-dependent process, the magnitude of the effect changes with the size of the population. Such processes are important for population stability.

The preparation of life tables often requires organisms to be aged. For woody plants this can be done from the growth rings in the trunk. For many animals, growth rings in hard structures can also be used: for example, fish are aged by the seasonal growth checks visible on the scales or otoliths. Similarly, molluscs can show growth bands in the shell. Damage and wear can be used to estimate the age of insects and mammals. For example, the development of and wear on teeth can be used to age mammals. Care must be taken when using growth checks, as these may also be caused by irregular events; for example, a drought or reproduction may also check growth.

Species Richness and Diversity

A key objective of ecologists is to estimate the biodiversity of habitats. Diversity can be used to assess the relative quality of different areas and as part of a long-term monitoring plan to ensure that a habitat is not degraded or harmed. The simplest measure of diversity is the total species number, but in most situations this can only be estimated as it would be prohibitively expensive to identify every rare species present. A number of computational techniques have been developed to estimate the total species richness from a series of samples. Another approach is to create a diversity index: this combines information on the numbers of species recorded with their relative abundance to produce a single number that can be used to track and compare the diversity of ecosystems.

Numerous diversity indices are used and no consensus exists as to the most appropriate one.

Recognizing Community Structure

Multivariate analysis is used when the objective is to search for relationships between or to classify samples that are defined by a number of attributes. These attributes may be the number of each species present or physical characteristics such as slope, nutrient content or pH. Data sets can be large: tropical forest beetle or tree studies can easily require analysis of a matrix of 100 samples (stations) by 350 species, and thus multivariate analysis requires a computer. Ordination techniques such as principal component analysis and correspondence analysis are frequently used

to search for relationships between sites. These approaches seek to present graphically the relationship in terms of faunal composition between the samples. Another approach is to use classificatory techniques that may present relationships using a dendrogram. Techniques such as two-way indicator species analysis combine aspects of both approaches to classify communities and recognize key species that can be used to identify each community.

References

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